

DIMITRIOS LEFTHERIOTIS

Undergraduate Student / Researcher · Department of Informatics, Ionian University

Corfu, Greece · Specializing in Applied AI & Computational Modelling

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ACADEMIC PROFILE & RESEARCH DIRECTION

Informatics undergraduate at **Ionian University** with strong foundations in algorithm engineering, computational complexity, and applied machine learning. Actively engaged in empirical AI research and evaluation of efficiency and sustainability of code generation. Practical background in local AI systems, athletic technology applications, and drone engineering.

EDUCATION

B.Sc. in Informatics (Candidate) — Ionian University

2024 – Present

- **Core Focus:** Algorithms & Data Structures, Machine Learning, Computational Modeling, Scientific Computing, Systems Architecture.
- **Institutional Context:** Department of Informatics, Corfu, Greece — Collaborating within departmental research initiatives (CMOD Lab).

RESEARCH & KEY ENGINEERING PROJECTS

1. Agentic Harness with Ollama

Lightweight, transparent AI coding agent harness for local execution with Ollama and compact models (7B/14B). Features multi-strategy tool parsing, 3-tier security gates, context pruning, and loop protection.

Repository: github.com/DimitriosLeftheriotis/Agentic-Harness-with-Ollama | **Stack:** Python, Ollama, Local LLMs, Tool Calling, CLI, Sandboxing

2. Pathfinding Visualizer

Interactive 2D grid pathfinding and graph traversal visualizer built with Pygame. Animates A*, Dijkstra, BFS, and DFS with real-time obstacle placement and frontier tracking.

Repository: github.com/DimitriosLeftheriotis/Pathfinding-Visualizer | **Stack:** Python, Pygame, A* Search, Dijkstra, BFS & DFS, Graph Algorithms

3. Athletic Training Periodization & Monitoring System (Annual Training Plan)

Flask web application converting coach-provided annual training data into interactive visual plans of intensity, volume, training periods, tests, and competitions.

Repository: github.com/DimitriosLeftheriotis/Annual-Training-Plan | **Stack:** Python, Flask, Matplotlib, NumPy, HTML, JavaScript

4. Empirical Energy & Computational Complexity Benchmarking of AI Code

Empirical investigation and statistical profiling evaluating the energy consumption, runtime efficiency, and sustainability of AI-assisted code generation.

Tools & Methods: Python, CodeCarbon, Matplotlib, Seaborn, Statistical Profiling

TECHNICAL COMPETENCIES

Programming & Systems:	Python, Java, C, C++, TypeScript, JavaScript, Flask, Pygame, HTML, XML, SQL, Bash, PowerShell, Linux, Git / GitHub, Docker
Machine Learning & AI:	Ollama, Local LLMs, Tool Calling, NumPy, Pandas, Scikit-Learn, Graph Algorithms
Profiling & Scientific Computing:	CodeCarbon, Matplotlib, Seaborn, Statistical Profiling
Hardware & Embedded:	Telemetry Calibration, Microcontroller Firmware (Arduino / ESP32), Drone Flight Controllers